

Health Risk Prediction Using Machine Learning Across Heterogeneous Patient Populations



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Motivation

- *ICU decisions must be made quickly under pressure*
- *Incorrect predictions can affect patient outcomes*
- *Patient data is complex and varies across cases*



Problem

<i>Problem Context</i>	<i>Real-Word Issue</i>
<i>ICU mortality prediction relies on complex clinical data</i>	<i>ICU data is distributed across multiple tables</i>
<i>Multiple features must be combined for accurate modeling</i>	<i>Missing values and variability increase modeling difficulty</i>

Objective

01

Predict ICU patient mortality using classification models

02

Combine clinical features into a unified input

03

Prepare and structure data for modeling

04

Address missing values and data variability

05

Evaluate models using performance metrics

Dataset

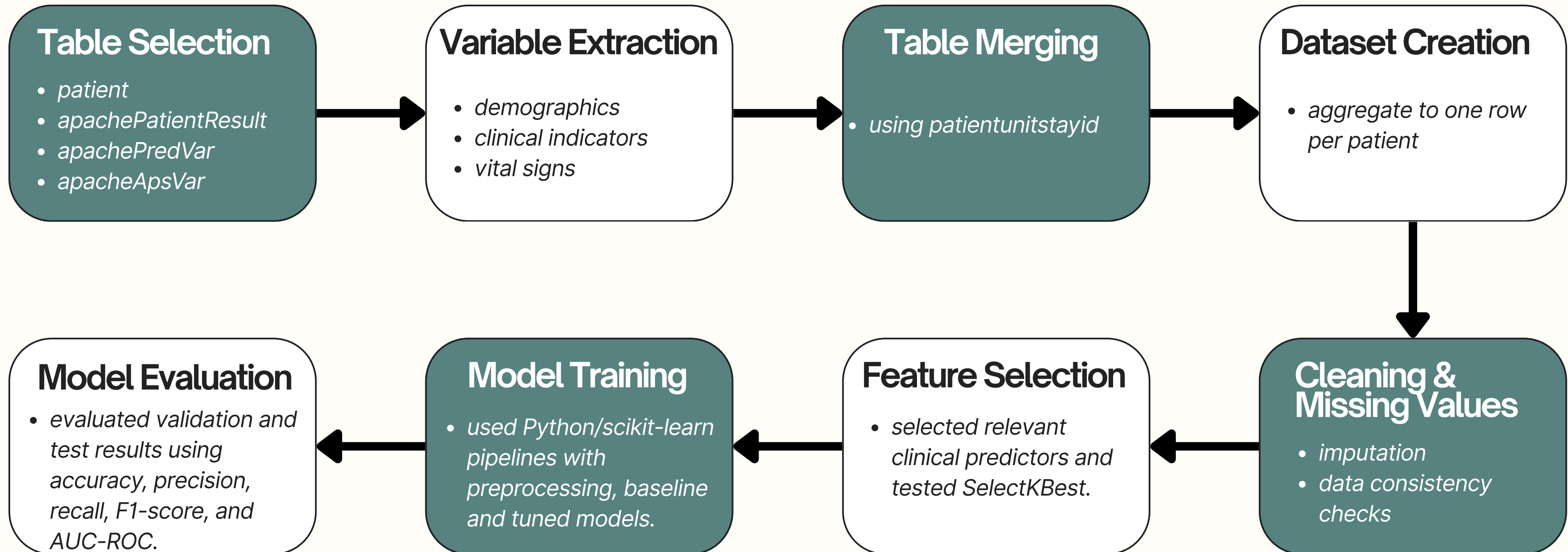
- *eICU Collaborative Research Database (PhysioNet)*
- *Multi-center ICU dataset with de-identified patient records*
- *Contains vital signs, diagnoses, treatments, and severity scores (e.g., APACHE)*
- *Includes data from thousands of ICU stays across multiple hospitals (only a subset were used)*
- *Final prepared dataset contained 2,520 ICU patient records.*



Key Challenge

- *Target was imbalanced: 2,308 survived and 212 died.*
- *This imbalance made mortality prediction harder because the model saw far fewer positive mortality cases.*

Pipeline



Method

Approach

- Binary classification (ICU mortality prediction)
- **Input:** processed clinical features
- **Output:** patient outcome (alive / deceased)
- Data preprocessing and feature engineering applied before modeling

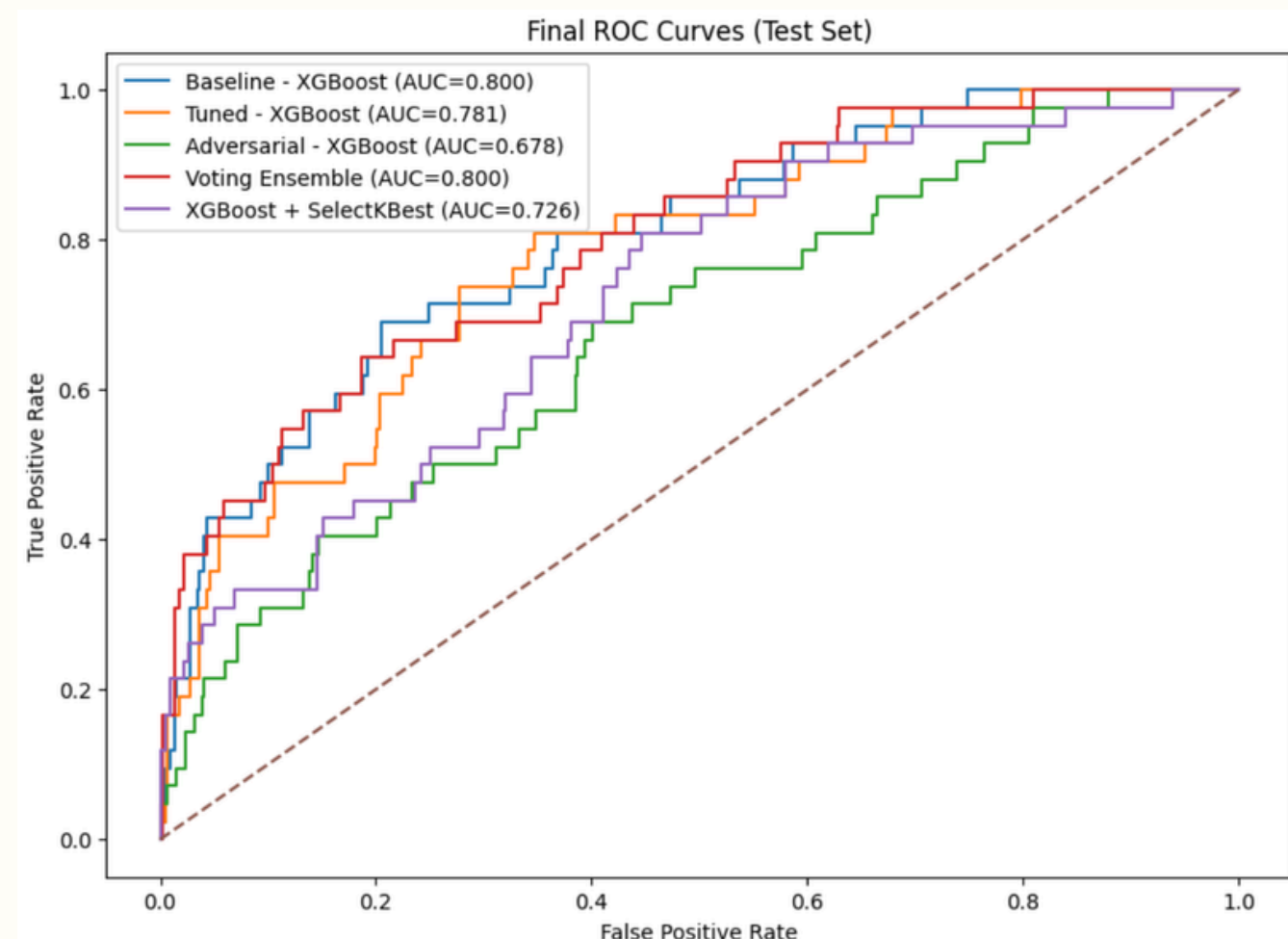
Models Tested

- Logistic Regression, Random Forest, Decision Tree
- KNN, SVM, Naive Bayes
- Gradient Boosting, XGBoost
- Models implemented using scikit-learn and XGBoost libraries

Model Selection, Tuning & Ensemble

- Selected top-performing models: Logistic Regression, XGBoost, Gradient Boosting
- Hyperparameter tuning performed using GridSearchCV with 5-fold cross-validation
- **Key tuned parameters:**
 - XGBoost: learning rate, max depth, number of estimators
 - Gradient Boosting: learning rate, number of estimators, max depth
 - Logistic Regression: regularization strength (C)
- **Final model:** Voting Ensemble
- Ensemble improved performance, especially for the minority class (higher F1-score)

Evaluation & Results



- *XGBoost was selected as a strong base model and used in the final ensemble*
- *Improving performance on the minority class was challenging*
- *Initial F1-score was very low (~0.1)*
- *Applied hyperparameter tuning and ensemble learning to improve performance*
- *Tested adversarial training by introducing controlled noise/perturbations to improve model robustness*
- *Performance improved, but class imbalance remains a limitation*
- *High accuracy with lower F1-score indicates class imbalance*

	Model	Accuracy	Precision	Recall	F1-score	AUC-ROC
3	Voting Ensemble	0.892857	0.380000	0.452381	0.413043	0.800041
0	Baseline - XGBoost	0.797619	0.227273	0.595238	0.328947	0.799732
1	Tuned - XGBoost	0.861111	0.274194	0.404762	0.326923	0.780561
4	XGBoost + SelectKBest	0.815476	0.200000	0.404762	0.267717	0.725675
2	Adversarial - XGBoost	0.561508	0.125523	0.714286	0.213523	0.678056

Reflection & Future Work

Reflection

- *Class imbalance significantly impacted model performance*
- *Accuracy was misleading because the dataset was imbalanced; F1-score was more informative because it combines precision and recall for the minority mortality class.*
- *Real-world clinical data is complex and noisy*
- *Model improvements required multiple strategies*

Future Work

- *Apply class balancing techniques (e.g., SMOTE, class weighting)*
- *Improve feature engineering for better signal extraction*
- *Explore more advanced ensemble methods*
- *Focus on improving minority class prediction*

Conclusion

- Developed an ICU mortality prediction pipeline using clinical data
- Compared multiple models and identified top performers
- Voting Ensemble achieved the best overall performance
- Model performance improved through tuning and model selection
- Results highlight challenges of predicting the minority class



Thank You For Listening!

Any Questions?

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